Администрирование сетевых подсистем

Расширенные настройки межсетевого экрана (Лабораторная работа №7)

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Цели и задачи работы

Цель лабораторной работы

Получить практические навыки настройки межсетевого экрана Firewalld в Linux, включая переадресацию портов и активацию Masquerading.

Выполнение лабораторной работы

Создание пользовательской службы Firewalld

root@server:/etc/firewalld/services — sudo -i

```
[manovichkov@server.manovichkov.net ~]$ sudo -i
[sudo] password for manovichkov:
[root@server.manovichkov.net ~]# cp /usr/lib/firewalld/services/ssh.xml
cp: missing destination file operand after '/usr/lib/firewalld/services/ssh.xml'
Try 'cp --help' for more information.
[root@server.manovichkov.net ~]# cp /usr/lib/firewalld/services/ssh.xml /etc/firewalld/services/ssh-custom.xml
[root@server.manovichkov.net ~]# cd /etc/firewalld/services/
[root@server.manovichkov.net services]# cat ssh-custom.xml
<?xml version="1.0" encoding="utf-8"?>
<service>
  <short>SSH</short>
  <description>Secure Shell (SSH) is a protocol for logging into and executing commands on remote machines. It provides se
cure encrypted communications. If you plan on accessing your machine remotely via SSH over a firewalled interface, enable
this option. You need the openssh-server package installed for this option to be useful.</description>
  <port protocol="tcp" port="22"/>
</service>
[root@server.manovichkov.net services]#
```

Рис. 1: Создание пользовательской службы SSH

Проверка служб Firewalld

```
[root@server.manovichkov.net services]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 anno-1800 apcupsd
aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-storage bb bgp bitcoin bitcoin-rpc
 bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter ceph-mon cfengine checkmk-agent civilization-iv civ
ilization-v cockpit collectd condor-collector cratedb ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc
dns dns-over-quic dns-over-tls docker-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio
 finger foreman foreman-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-cl
ient ganglia-master git gpsd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 ipfs ipp ipp-cl
ient ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpasswd kprop kshell kube-api kube-
apiserver kube-control-plane kube-control-plane-secure kube-controller-manager kube-controller-manager-secure kube-nodepor
t-services kube-scheduler kube-scheduler-secure kube-worker kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt lib
virt-tls lightning-network llmnr llmnr-client llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp
 mongodb mosh mountd mpd mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-
dashboard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmconsole plex pm
cd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporter proxy-dhcp ps2link ps3nets
rv ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel rootd rpc-bind rquotad rsh rsyncd rtsp salt-
master samba samba-client samba-dc sane settlers-history-collection sip sips slimevr slp smtp smtp-submission smtps snmp s
nmptls snmptls-trap snmptrap spideroak-lansync spotify-sync squid ssdp ssh ssh-custom statsrv steam-lan-transfer steam-str
eaming stellaris stronghold-crusader stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay
synergy syscomlan syslog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns upn
p-client vdsm vnc-server vrrp warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-discovery-host
 ws-discovery-tcp ws-discovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client xmpp-local xmpp-server zabbix-a
gent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service zero-k zerotier
[root@server.manovichkov.net services]#
```

Перенаправление портов

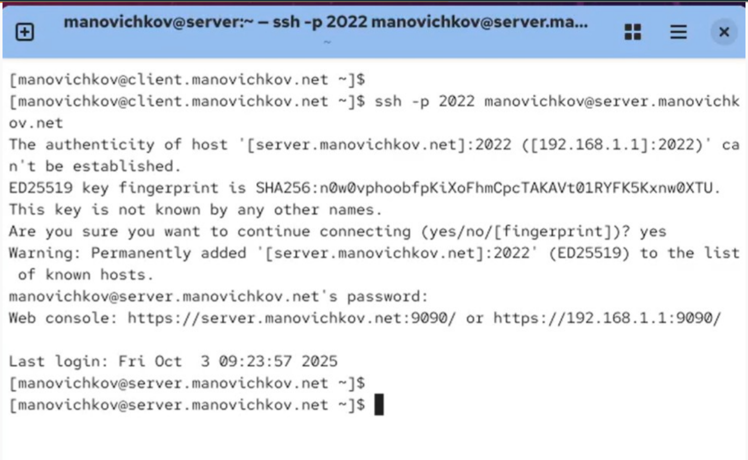

Рис. 3: Подключение к серверу по SSH через порт 2022

Настройка Port Forwarding и Masquerading

```
[root@server.manovichkov.net services]# echo "net.ipv4.ip_forward = 1" > /etc/sysctl.d/90-forward.conf
[root@server.manovichkov.net services]# sysctl -p /etc/sysctl.d/90-forward.conf
net.ipv4.ip_forward = 1
[root@server.manovichkov.net services]# firewall-cmd --zone=public --add-masquerade --permanent
success
[root@server.manovichkov.net services]# firewall-cmd --reload
success
[root@server.manovichkov.net services]#
```

Рис. 4: Включение IPv4 forwarding и маскарадинга

Проверка доступа в Интернет

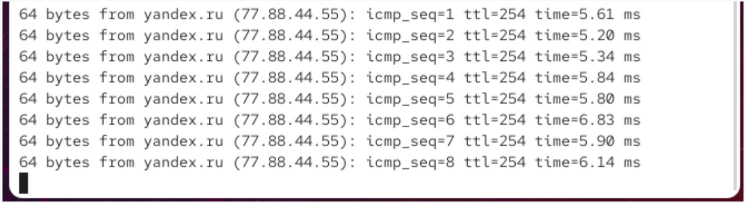

Рис. 5: Проверка доступа к Интернету с клиента

Подготовка окружения и автоматизация

```bash
#!/bin/bash
echo "Provisioning script $0"
echo "Copy configuration files"
cp -R /vagrant/provision/server/firewall/etc/* /etc
echo "Configure masquerading"
firewall-cmd --add-service=ssh-custom --permanent
firewall-cmd --add-forward-port=port=2022:proto=tcp:toport=22 --permanent
firewall-cmd --zone=public --add-masquerade --permanent
firewall-cmd --reload
restorecon -vR /etc
```

Рис. 6: Содержимое скрипта firewall.sh

Выводы по проделанной работе

Вывод

- Настроен пользовательский сервис Firewalld с новым портом SSH.
- Реализовано перенаправление портов и NAT.
- Активировано IPv4-перенаправление и маскарадинг.
- Создан скрипт автоматизации, упрощающий последующее развёртывание конфигурации.